Preliminary Numerical Evidence for Universal Anti-Correlation between Hadamard Amplitudes and Minimum Zero Spacing across Self-Dual and Non-Self-Dual L-Functions

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Abstract

For a degree-d L-function $L(s)$ in the Selberg class with simple zeros $\rho_n = \frac{1}{2} + i\gamma_n$ on the critical line, the *Hadamard amplitude*

$$A_\Lambda(\gamma_n) \;=\; \mathrm{Im}(c_0)^2 + 2\,\mathrm{Re}(c_1),$$

extracted from the Laurent expansion $\Lambda'/\Lambda(\rho_n + u) = u^{-1} + c_0 + c_1 u + \cdots$, measures the local spectral weight of each zero. We present numerical evidence that the Spearman rank anti-correlation $\rho_{\mathrm{S}}(A_\Lambda,\, g_{\min}^{\mathrm{GUE}})$ is *universal* across ten automorphic L-functions of degree 1 through 4 and conductor 1 through 1,331, encompassing both self-dual ($\varepsilon = \pm 1$) and non-self-dual ($\varepsilon \in S^1$) root numbers. The overall mean correlation is $\rho_{\mathrm{S}} = -0.892 \pm 0.015$ (range 0.043). The minimum gap $g_{\min}$ is identified as the canonical gap metric via height-stability analysis (span 0.16 across eight height bins, $p = 0.40$), in contrast to g_{right} which decays systematically with height. These results constitute preliminary evidence for a general universality of the amplitude–gap anti-correlation across the Langlands hierarchy, within the tested range of degree and conductor.

1 Introduction

The distribution of nontrivial zeros of L-functions has been a central object of study since Riemann's 1859 memoir. While the vertical distribution—zero density, pair correlation—is well understood through the work of Montgomery [1], Odlyzko [2], and Rudnick–Sarnak [3], the *individual* character of each zero has received comparatively little attention.

In the companion papers [4, 5, 6], we developed the ξ-bundle framework, in which the completed L-function $\Lambda(s)$ defines a $U(1)$-connection on the critical line with curvature $\kappa = |\Lambda'/\Lambda|^2$. Near a simple zero $\rho_n = \frac{1}{2} + i\gamma_n$, the curvature admits the Laurent expansion $\kappa\,\delta^2 = 1 + A(\gamma_n)\,\delta^2 + O(\delta^3)$, where $\delta = |s - \rho_n|$ and the coefficient $A(\gamma_n) = \mathrm{Im}(c_0)^2 + 2\,c_1$ is the *Hadamard amplitude* of the zero.

Discovery trajectory. The anti-correlation between A and the local zero spacing was first observed for the Riemann zeta function in Result C-270, where $\rho_{\mathrm{S}}(A_\Lambda, g_{\min}^{\mathrm{GUE}}) = -0.900$ ($n = 910$ zeros, $\gamma \le 2000$). Result C-375 extended the measurement to five self-dual L-functions of degree 1–3 (conductor 1–121), finding $\rho_{\mathrm{S}} = -0.874 \pm 0.028$ across all five.

Result C-378 further extended to four non-self-dual L-functions with complex root numbers $\varepsilon \in S^1$, yielding $\rho_S = -0.887 \pm 0.015$. Result C-383 extended the measurement to degree 4 via $L(s, \mathrm{Sym}^3 E_{11a1})$, obtaining $\rho_S = -0.905$ ($n = 327$), consistent with the degree 1–3 average. The combined ten-family dataset is the subject of the present paper.

Main results.

1. **Universal anti-correlation** (Section 4): ten L-functions of degree 1–4 give $\rho_S(A_\Lambda, g_{\min}^{\mathrm{GUE}}) = -0.892 \pm 0.015$ (range 0.043), with self-dual mean -0.895 and non-self-dual mean -0.887.

2. **Canonical metric identification** (Section 4.2): $g_{\min}$ is height-stable (span 0.16, $p = 0.40$) while g_{right} decays with height (span 0.57, $p = 5 \times 10^{-5}$).

3. **Root-number independence** (Section 4.3): the complex root number $\varepsilon \in S^1$ does not affect the correlation ($\Delta\rho = 0.006$, well within confidence intervals), ruling out self-duality as a necessary condition.

Scope and limitations. All results are numerical, covering degree 1–4 and conductor 1–1,331 only. Maass forms, $\mathrm{GL}(n)$ L-functions with $n \geq 4$, and high-conductor regimes remain untested. We therefore describe our findings as *preliminary numerical evidence for general universality*, following the classification of the companion papers.

2 Framework

2.1 Selberg class and completed L-functions

Let $L(s)$ belong to the Selberg class $\mathcal{S}$ with degree d, conductor N, and root number ε ($|\varepsilon| = 1$). The completed L-function is

$$\Lambda(s) = \gamma(s)\, L(s), \qquad \Lambda(s) = \varepsilon\, \overline{\Lambda(1 - \bar{s})}, \tag{1}$$

where $\gamma(s) = \prod_{j=1}^{d} \Gamma_{\mathbb{R}}(s + \mu_j)$ is the Gamma factor with spectral parameters μ_j.

For self-dual L-functions, $\varepsilon = \pm 1$ and the zeros are symmetric: $\gamma \in \mathrm{zeros}(L)$ implies $-\gamma \in \mathrm{zeros}(L)$. For non-self-dual L-functions (e.g., Dirichlet L-functions with complex primitive characters), $\varepsilon \in S^1 \setminus \{\pm 1\}$ and $\gamma \in \mathrm{zeros}(L(s, \chi))$ implies $-\gamma \in \mathrm{zeros}(L(s, \bar{\chi}))$—the negative-height zeros belong to the conjugate L-function.

2.2 Hadamard amplitude A_Λ

At a simple zero $\rho_n = \frac{1}{2} + i\gamma_n$, the logarithmic derivative has a simple pole:

$$\frac{\Lambda'}{\Lambda}(\rho_n + u) = \frac{1}{u} + c_0 + c_1\, u + c_2\, u^2 + \cdots . \tag{2}$$

By the parity theorem ([5, Theorem 5]), $c_0 \in i\mathbb{R}$ and $c_1 \in \mathbb{R}$ for on-critical zeros. Setting $B = \mathrm{Im}(c_0)$ and $H_1 = c_1$:

Definition 2.1 (Hadamard amplitude)**.**

$$A_\Lambda(\gamma_n) \;=\; B(\gamma_n)^2 + 2\,H_1(\gamma_n) \;=\; \mathrm{Im}(c_0)^2 + 2\,c_1. \tag{3}$$

The Hadamard product representation yields the decomposition:

$$B(\gamma_n) = -\sum_{k\neq n}{}' \frac{1}{\gamma_n - \gamma_k} + \Gamma\text{-terms}, \tag{4}$$

$$H_1(\gamma_n) = \sum_{k\neq n}{}' \frac{1}{(\gamma_n - \gamma_k)^2} + \Gamma'\text{-terms}. \tag{5}$$

Since every term in the H_1 sum (5) is positive, $H_1 > 0$ unconditionally. The sum B is alternating and partially cancellative.

Theorem 1 (Amplitude–gap lower bound, [7, Theorem 1])**.** *Let $L(s) \in \mathcal{S}$ have simple zeros on the critical line. Then*

$$A_\Lambda(\gamma_n) \;\geq\; \frac{2}{\Delta_{\min}(\gamma_n)^2}, \tag{6}$$

where $\Delta_{\min} = \min(\gamma_n - \gamma_{n-1},\ \gamma_{n+1} - \gamma_n)$.

Proof. $A_\Lambda = B^2 + 2H_1 \geq 2H_1 \geq 2(1/\Delta_L^2 + 1/\Delta_R^2) \geq 2/\Delta_{\min}^2$. □

This bound is unconditional and height-independent, relying only on the Hadamard product structure. Numerically, the nearest-neighbour contribution $2H_1^{\mathrm{NN}}/A_\Lambda \approx 87\%$ for $\zeta(s)$, so the bound captures the dominant mechanism.

2.3 GUE-normalised gap metrics

The theoretical zero density at height t for an L-function of degree d and conductor N is:

$$\bar{d}(t) = \frac{1}{2\pi}\left[d\,\log\!\left(\frac{t}{2\pi}\right) + \log N\right] + (\text{Gamma corrections}). \tag{7}$$

Definition 2.2 (Gap metrics)**.** The GUE-normalised gap metrics for the n-th zero are:

$$g_{\min}^{\mathrm{GUE}}(\gamma_n) = \min(\gamma_n - \gamma_{n-1},\ \gamma_{n+1} - \gamma_n)\cdot \bar{d}(\gamma_n), \tag{8}$$

$$g_{\mathrm{right}}^{\mathrm{GUE}}(\gamma_n) = (\gamma_{n+1} - \gamma_n)\cdot \bar{d}(\gamma_n). \tag{9}$$

The density $\bar{d}(t)$ in (7) is the *theoretical* density from the explicit formula, not the empirical estimate $2/(\gamma_{n+1} - \gamma_{n-1})$. This distinction is critical: the empirical estimate introduces spurious correlations between the gap and the normalisation factor (cf. gap normalisation artefacts documented in [5]).

3 Data: Nine L-Function Families

We measure the Spearman correlation $\rho_{\mathrm{S}}(A_\Lambda, g_{\min}^{\mathrm{GUE}})$ for ten L-functions spanning four degrees, conductors from 1 to 1,331, and both self-dual and non-self-dual root numbers.

Computational protocol. For each L-function:

Table 1: Ten L-functions and their amplitude–gap correlations under the canonical $(A_\Lambda, g_{\min}^{\mathrm{GUE}})$ metric. The five self-dual families of degree 1–3 are from Result C-375; the degree-4 family is from Result C-383; the four non-self-dual families are from Result C-378. All p-values are below 10^{-24}.

L-function	d	N	n	ρ_{S}	95% **CI**	**Root #**	**Type**
$\zeta(s)$	1	1	910	-0.900	$[-0.915, -0.882]$	$+1$	self-dual
$L(s, \chi_{-3})$	1	3	494	-0.891	$[-0.908, -0.871]$	-1	self-dual
$L(s, E_{11\mathrm{a}1})$	2	11	437	-0.914	$[-0.928, -0.897]$	$+1$	self-dual
$L(s, \Delta)$	2	1	324	-0.876	$[-0.899, -0.848]$	$+1$	self-dual
$L(s, \mathrm{sym}^2 E_{11})$	3	121	96	-0.885	$[-0.920, -0.830]$	$+1$	self-dual
$L(s, \mathrm{Sym}^3 E_{11})$	4	1331	327	-0.905	$[-0.928, -0.873]$	$+1$	self-dual
$L(s, \chi_5^{(1)})$	1	5	238	-0.903	$[-0.929, -0.866]$	$\varepsilon \in S^1$	non-self-dual
$L(s, \bar{\chi}_5^{(1)})$	1	5	238	-0.875	$[-0.910, -0.828]$	$\bar{\varepsilon}$	non-self-dual
$L(s, \chi_7^{(1)})$	1	7	254	-0.871	$[-0.908, -0.820]$	$\varepsilon \in S^1$	non-self-dual
$L(s, \bar{\chi}_7^{(1)})$	1	7	255	-0.901	$[-0.927, -0.866]$	$\bar{\varepsilon}$	non-self-dual
Self-dual mean (6 families)				-0.895			
Non-self-dual mean (4 families)				-0.887			
Overall mean (10 families)				$\mathbf{-0.892}$			
Overall std				0.015			
Overall range				0.043			

1. Collect all zeros γ_n up to height T via PARI/GP (`lfunzeros`).
2. Compute $A_\Lambda(\gamma_n)$ from the full Hadamard zero-sum with Gamma corrections derived from the spectral parameters μ_j.
3. Normalise gaps by the theoretical density $\bar{d}(t)$ from (7).
4. Trim the outermost 20% on each side (central 60%) to avoid boundary effects.
5. Report Spearman ρ_{S} with bootstrap 95% confidence intervals (2000 resamples).

For non-self-dual characters $\chi \pmod q$, the Hadamard sum over negative-height zeros uses the zeros of $\bar{\chi}$ ("proper" method). The "naïve" method (using χ's own zeros for the conjugate sum) yields $|\Delta\rho| < 0.0003$, confirming that the conjugate-sum contribution is a smooth, position-insensitive baseline.

Table 1 presents the complete dataset. The ten families span:

- **Degree**: $d = 1$ (Riemann ζ, Dirichlet), $d = 2$ (elliptic curve $E_{11\mathrm{a}1}$, Ramanujan Δ), $d = 3$ (symmetric square $\mathrm{sym}^2 E_{11}$), $d = 4$ (symmetric cube $\mathrm{Sym}^3 E_{11}$).
- **Conductor**: $N = 1, 3, 5, 7, 11, 121, 1331$.
- **Root number**: $\varepsilon = +1, -1$ (self-dual) and $\varepsilon \in S^1 \setminus \{\pm 1\}$ (non-self-dual, $\varepsilon = 0.851 + 0.526i$ for mod 5, $\varepsilon = 0.387 + 0.922i$ for mod 7).
- **Total zeros**: $\sum n_i = 3{,}573$ interior zeros (after 20% trimming on each side).

4 Results

4.1 Universality of the anti-correlation

All ten L-functions in Table 1 exhibit a strong negative Spearman correlation $\rho_S(A_\Lambda, g_{\min}^{\mathrm{GUE}})$:

- Every family satisfies $\rho_S < -0.87$.
- The overall mean is -0.892 ± 0.015 (std).
- The range is 0.043 (-0.914 for E_{11a1} to -0.871 for $\chi_7^{(1)}$).
- Degree dependence is absent: degree 1 mean $= -0.890$ ($n = 6$), degree 2 mean $= -0.895$ ($n = 2$), degree 3 $= -0.885$ ($n = 1$), degree 4 $= -0.905$ ($n = 1$).
- Conductor dependence is absent over a 1,331-fold range ($N = 1$ to 1,331).

The numerical evidence suggests that the anti-correlation is a universal property of L-functions in the tested range, not an artefact of a specific family, degree, or root number.

4.2 $g_{\min}$ as the canonical gap metric

To determine which gap metric is intrinsic to the amplitude anti-correlation, we partition the 910 interior zeros of $\zeta(s)$ ($\gamma \leq 2000$) into eight height bins and measure the correlation in each (Table 2).

Table 2: Height-bin analysis for $\zeta(s)$ ($n = 910$, eight bins). The $(A_\Lambda, g_{\min})$ pairing is height-stable (span 0.163, $p = 0.40$), while $(A_\Lambda, g_{\mathrm{right}})$ decays systematically.

Bin	$\bar{t}$	$\rho(A_\Lambda, g_{\min})$	$\rho(A_\Lambda, g_{\mathrm{right}})$	n
1	626	-0.814	-0.686	114
2	778	-0.889	-0.641	114
3	924	-0.938	-0.624	113
4	1064	-0.961	-0.504	114
5	1202	-0.977	-0.496	114
6	1337	-0.948	-0.339	113
7	1469	-0.954	-0.262	114
8	1599	-0.867	-0.120	114
Span		0.163	0.567	
Kendall τ		-0.29	$+1.00$	
p-value		0.40	5×10^{-5}	

Observation 2 ($g_{\min}$ as canonical metric)**.** *The pairing* $(A_\Lambda,\ g_{\min}^{\mathrm{GUE}})$ *is the unique height-stable combination among the four natural pairings* (A_Λ *or* A_L *with* $g_{\min}$ *or* g_{right})*. Its span of* 0.163 *is below the* 0.20 *threshold, and the Kendall trend is not significant* ($p = 0.40$)*. All three other pairings exhibit significant monotonic trends* ($p < 0.01$)*.*

The mechanism for height stability is the unconditional bound (6): since $A_\Lambda \geq 2/\Delta_{\min}^2$ is purely algebraic and height-independent, and since the nearest-neighbour contribution accounts for ∼87% of A_Λ, the rank correlation inherits the height-independence of the bound. The g_{right} metric, by contrast, receives Gamma-factor contamination that grows as $\sim \log t$.

4.3 Non-self-dual L-functions

Four non-self-dual L-functions—two conjugate pairs $(\chi_5^{(1)}, \bar{\chi}_5^{(1)})$ and $(\chi_7^{(1)}, \bar{\chi}_7^{(1)})$ with complex root numbers $\varepsilon \in S^1 \setminus \{\pm 1\}$—yield ρ_{S} values indistinguishable from the self-dual families:

Observation 3 (Root-number independence)**.** *Self-dual mean* $\rho_{\mathrm{S}} = -0.893$*; non-self-dual mean* $= -0.887$*. The difference* $\Delta = 0.006$ *is well within the individual* 95% *confidence interval widths (*$\sim$0.05–0.09*).*

This rules out the hypothesis that the self-dual constraint $\varepsilon = \pm 1$ is a necessary condition for the amplitude–gap anti-correlation.

Observation 4 (Conjugate-pair consistency)**.** *Within each conjugate pair:* $|\rho(\chi) - \rho(\bar{\chi})| = 0.029$ *(mod* 5*) and* 0.031 *(mod* 7*), with alternating sign. These differences are consistent with statistical fluctuation from independent zero sets of equal density.*

Remark 4.1. For non-self-dual characters, the "proper" Hadamard sum uses the zeros of $\bar{\chi}$ for the conjugate terms, while the "naïve" method uses χ's own zeros. The resulting $|\Delta\rho| < 0.0003$ in all four cases. The explanation is that the conjugate sum $\sum_k 1/(\gamma_n + \tilde{\gamma}_k)$ involves $\tilde{\gamma}_k \gg \gamma_n$ for most terms, making each contribution $\sim 1/(2\gamma_n)$, insensitive to the specific $\tilde{\gamma}_k$ values.

4.4 $g_{\min}$ versus g_{right}

Across all ten families, $g_{\min}$ consistently produces stronger correlations than g_{right}:

Metric	$\bar{\rho}_{\mathrm{S}}$ (ten families)	Height stability
$g_{\min}^{\mathrm{GUE}}$	-0.891	span 0.16, $p = 0.40$
$g_{\mathrm{right}}^{\mathrm{GUE}}$	-0.44 to -0.53	span 0.57, $p < 10^{-4}$

The $g_{\min}$ metric captures the symmetric nearest-neighbour structure predicted by the lower bound (6)—a zero with small minimum gap *must* have large amplitude, regardless of which side the close neighbour lies on.

5 Discussion

5.1 Theoretical context

The unconditional bound $A_\Lambda \geq 2/\Delta_{\min}^2$ (Theorem 1) guarantees that the anti-correlation sign is negative for any L-function with non-degenerate gap distribution. The *strength* of the correlation ($|\rho| \approx 0.89$) is not predicted by the bound alone; it reflects the tightness of the bound, which in turn depends on the dominance of the nearest-neighbour terms in the Hadamard sum.

The GUE random matrix ensemble produces $\rho_{\mathrm{S}}^{\mathrm{full}}(A, g_{\min}) = -0.912$ under matched conditions (full sum, 20% trim, $N = 200$–2000; see [7]). The observed -0.892 for the ten families is within 2% of this GUE baseline, suggesting that the bulk of the correlation is a consequence of the universal eigenvalue repulsion encoded in the GUE pair correlation function $R_2(x) = 1 - (\sin \pi x / \pi x)^2$.

5.2 Relation to the ξ-bundle framework

The amplitude A_Λ is the curvature coefficient in the ξ-bundle framework [4, 5]: the identity $\kappa\,\delta^2 - 1 = A_\Lambda\,\delta^2 + O(\delta^3)$ shows that A_Λ governs the leading deviation of the bundle curvature from its singular $1/\delta^2$ behaviour near a zero. The anti-correlation $\rho(A_\Lambda, g_{\min}) < 0$ therefore implies that zeros with small spacing exert stronger curvature influence—a "spectral influence radius" interpretation.

5.3 Limitations and untested regimes

The present evidence, while consistent across ten families, has clear limitations:

1. **Degree range**: $d = 1, 2, 3, 4$ have been tested, with degree 4 represented by a single L-function ($\mathrm{Sym}^3 E_{11a1}$, $N = 1{,}331$, $n = 327$, $\rho_{\mathrm{S}} = -0.905$). GL(5) and higher ($\mathrm{sym}^4$, Rankin–Selberg convolutions) remain unverified under the canonical $(A_\Lambda, g_{\min}^{\mathrm{GUE}})$ metric. Preliminary evidence from [7] using the uniform $(A_L, g_{\min})$ metric extends to degree 6 ($\mathrm{sym}^5 E_{11}$), but with weaker correlations ($\rho \approx -0.42$) that may reflect T-range limitations rather than a genuine degree effect.

2. **Conductor range**: $N \leq 1{,}331$. Higher-conductor effects (deeper Euler product structure, denser zero distribution) have not been extensively tested.

3. **Maass forms**: no Maass form L-function has been included. Maass forms have continuous spectral parameters, and the Gamma factor structure differs from holomorphic forms.

4. **Non-self-dual at $d \geq 2$**: the four non-self-dual families are all degree 1 (Dirichlet characters). Degree-2 non-self-dual L-functions (e.g., twisted elliptic curve L-functions, GL(2) Maass forms) have not been tested.

5. **Sample size**: the smallest family ($\mathrm{sym}^2 E_{11}$) has $n = 96$ interior zeros. While the 95% CI $[-0.920, -0.830]$ excludes -0.70 comfortably, larger samples would sharpen the estimate.

We emphasise that the consistency across ten families *suggests*, but does not *prove*, general universality. The mathematical content is the unconditional lower bound (Theorem 1); the universality claim rests on numerical evidence only.

6 Conclusion

We have presented numerical evidence that the Spearman anti-correlation $\rho_{\mathrm{S}}(A_\Lambda, g_{\min}^{\mathrm{GUE}})$ between the Hadamard amplitude and the GUE-normalised minimum gap is universal across ten automorphic L-functions of degree 1–4, conductor 1–1,331, with both self-dual and non-self-dual root numbers.

The key findings are:

1. The correlation $\rho_{\mathrm{S}} = -0.892 \pm 0.015$ (range 0.043) is remarkably stable across families, suggesting a universal mechanism rooted in the Hadamard product structure.

2. The minimum gap $g_{\min}$ is the canonical metric: it is the unique gap metric yielding a height-stable correlation with A_Λ.

3. Self-dual and non-self-dual families produce indistinguishable correlations ($\Delta = 0.006$), ruling out the root-number constraint as a necessary condition.

4. The observed correlation is within 2% of the GUE prediction (-0.912), consistent with the random-matrix universality of local zero statistics.

The unconditional lower bound $A_\Lambda \geq 2/\Delta^2_{\min}$ provides the theoretical underpinning: the anti-correlation sign is guaranteed by the positivity of the Hadamard sum H_1, and the correlation strength reflects the tightness of this bound ($2H_1^{\mathrm{NN}}/A \approx 87\%$).

Future directions include: extending to GL(4+) and Maass forms, testing height-bin stability within each non-ζ family, and investigating whether the $2H_1/A_L \approx 2/3$ ratio observed across all degrees can be derived analytically from the GUE pair correlation function.

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